03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 13

(JCD 8.5 - 9)

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Previous lecture

- Average reaction rate: $r_{aA} = n_A\, n_a\, \langle \sigma\, v \rangle$
- Cross section-velocity distribution:

$$\langle \sigma\, v \rangle = \left(\frac{8}{m\pi} \right)^{1/2} \frac{1}{(k\, T)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{k\, T} - \frac{b}{\sqrt{E}} \right) dE$$

- The Gamow peak is the narrow range of energies around the maximum of the exponential in this function.
- Energy generation rate: $\epsilon_{Aa} = \frac{r_{Aa}\, Q_{Aa}}{\rho}$
- Hydrogen fusion is summarised as $4\, {}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + 2\, e^+ + 2\, \nu_e$ where a total of 26.73 MeV is released

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Outline

Proton-proton chain

CNO cycle

Nucleosynthesis of heavier elements

Equations of stellar structure

Summary

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Proton-proton I chain

- Step 1 - average reaction time $\sim 10^9$ yr:

$$ {}^1_1\mathrm{H} + {}^1_1\mathrm{H} \rightarrow {}^2_1\mathrm{D} + e^+ + \nu $$

- Step 2 - average reaction time $< 10\,\mathrm{s}$:

$$ {}^2_1\mathrm{D} + {}^1_1\mathrm{H} \rightarrow {}^3_2\mathrm{He} + \gamma $$

- Step 3 - average reaction time $\sim 10^6$ yr:

$$ {}^3_2\mathrm{He} + {}^3_2\mathrm{He} \rightarrow {}^4_2\mathrm{He} + 2\,{}^1_1\mathrm{H} $$

- Effective energy release is $Q_{\mathrm{eff}} = 26.21\,\mathrm{MeV}$
- Total energy generation rate:

$$ \epsilon_{pp} \propto \frac{n_H^2\, T^4}{\rho} \propto X^2\, \rho\, T^4 $$

Proton-proton II chain

- Step 1 and 2 remain the same: ${}^1_1\mathrm{H}({}^1_1\mathrm{H}, e^+\,\nu){}^2_1\mathrm{D}({}^1_1\mathrm{H}, \gamma){}^3_2\mathrm{He}$
- And thus also the total energy generation rate
- Step 3
$$ {}^3_2\mathrm{He} + {}^4_2\mathrm{He} \rightarrow {}^7_4\mathrm{Be} + \gamma $$
- Step 4
$$ {}^7_4\mathrm{Be} + e^- \rightarrow {}^7_3\mathrm{Li} + \nu_e $$
- Step 5
$$ {}^7_3\mathrm{Li} + {}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + {}^4_2\mathrm{He} $$
- Effective energy release is $Q_{\mathrm{eff}} = 25.67\,\mathrm{MeV}$
- Happens in $\sim 30\%$ of the reactions

Proton-proton III chain

- Step 1, 2, and 3 from pp-II remain the same:

$$ {}^{1}_{1}\mathrm{H}({}^{1}_{1}\mathrm{H}, e^{+}\,\nu){}^{2}_{1}\mathrm{D}({}^{1}_{1}\mathrm{H}, \gamma){}^{3}_{2}\mathrm{He}({}^{4}_{2}\mathrm{He}, \gamma){}^{7}_{4}\mathrm{Be} $$

- And thus also the total energy generation rate
- Step 4

$$ {}^{7}_{4}\mathrm{Be} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{8}_{5}\mathrm{B} + \gamma $$

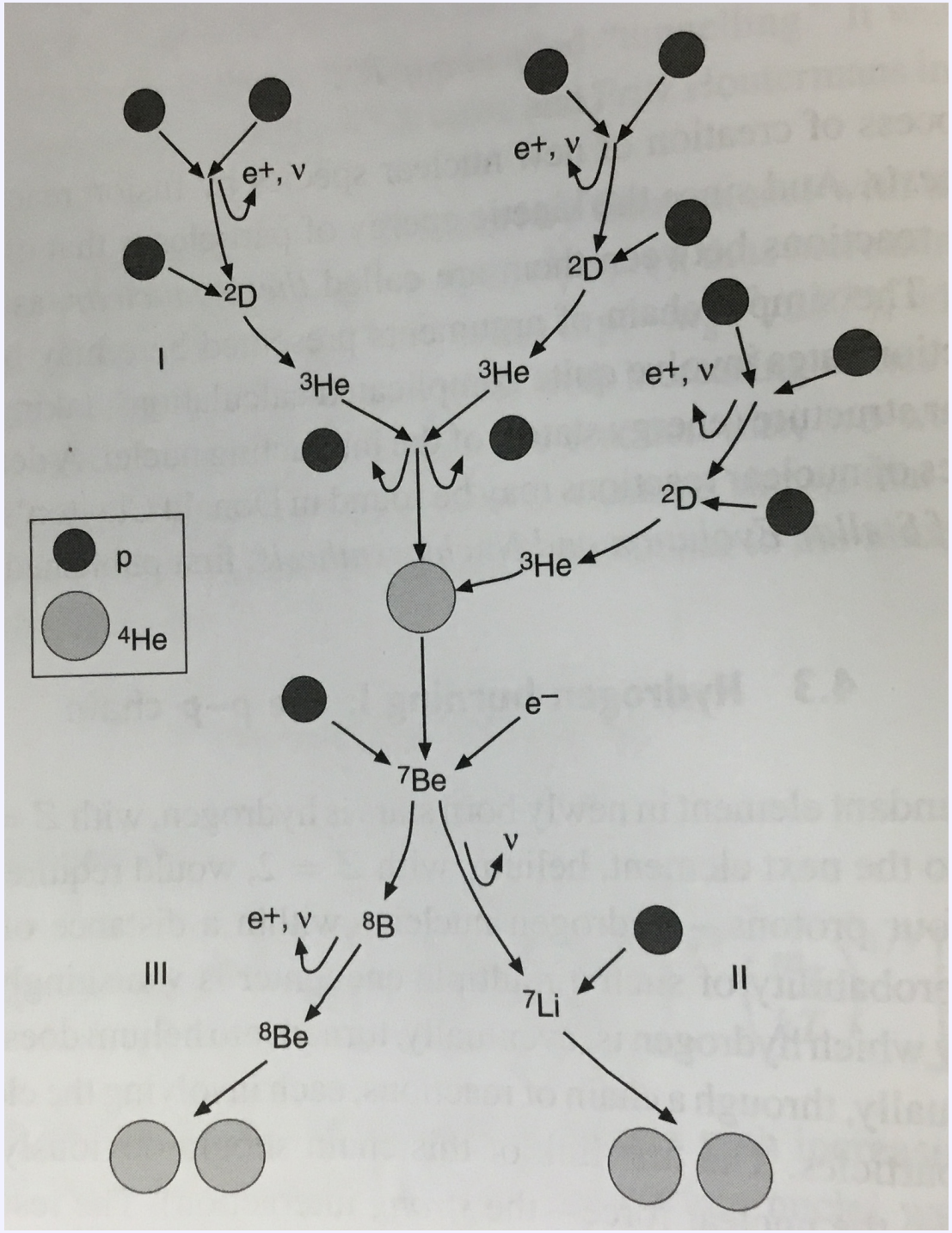
- Step 5

$$ {}^{8}_{5}\mathrm{B} \rightarrow {}^{8}_{4}\mathrm{Be} + e^{+} + \nu_{e} $$

- Step 6

$$ {}^{8}_{4}\mathrm{Be} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{4}_{2}\mathrm{He} $$

- Effective energy release is $Q_{\mathrm{eff}} = 19.27\,\mathrm{MeV}$
- Only happens rarely, but the reaction is important to try and detect Solar neutrino's



(Figure credit: Prialnik)

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CNO cycle - actually cyclic

- Step 1 - average reaction time $\sim 5 \cdot 10^3$ yr:

$$^{12}_{6}\mathrm{C} + ^{1}_{1}\mathrm{H} \rightarrow ^{13}_{7}\mathrm{N} + \gamma$$

- Step 2 - average reaction time ~ 15 min:

$$^{13}_{7}\mathrm{N} \rightarrow ^{13}_{6}\mathrm{C} + e^{+} + \nu$$

- Step 3 - average reaction time $\sim 2 \cdot 10^3$ yr:

$$^{13}_{6}\mathrm{C} + ^{1}_{1}\mathrm{H} \rightarrow ^{14}_{7}\mathrm{N} + \gamma$$

- Step 4 - average reaction time $\sim 10^6$ yr:

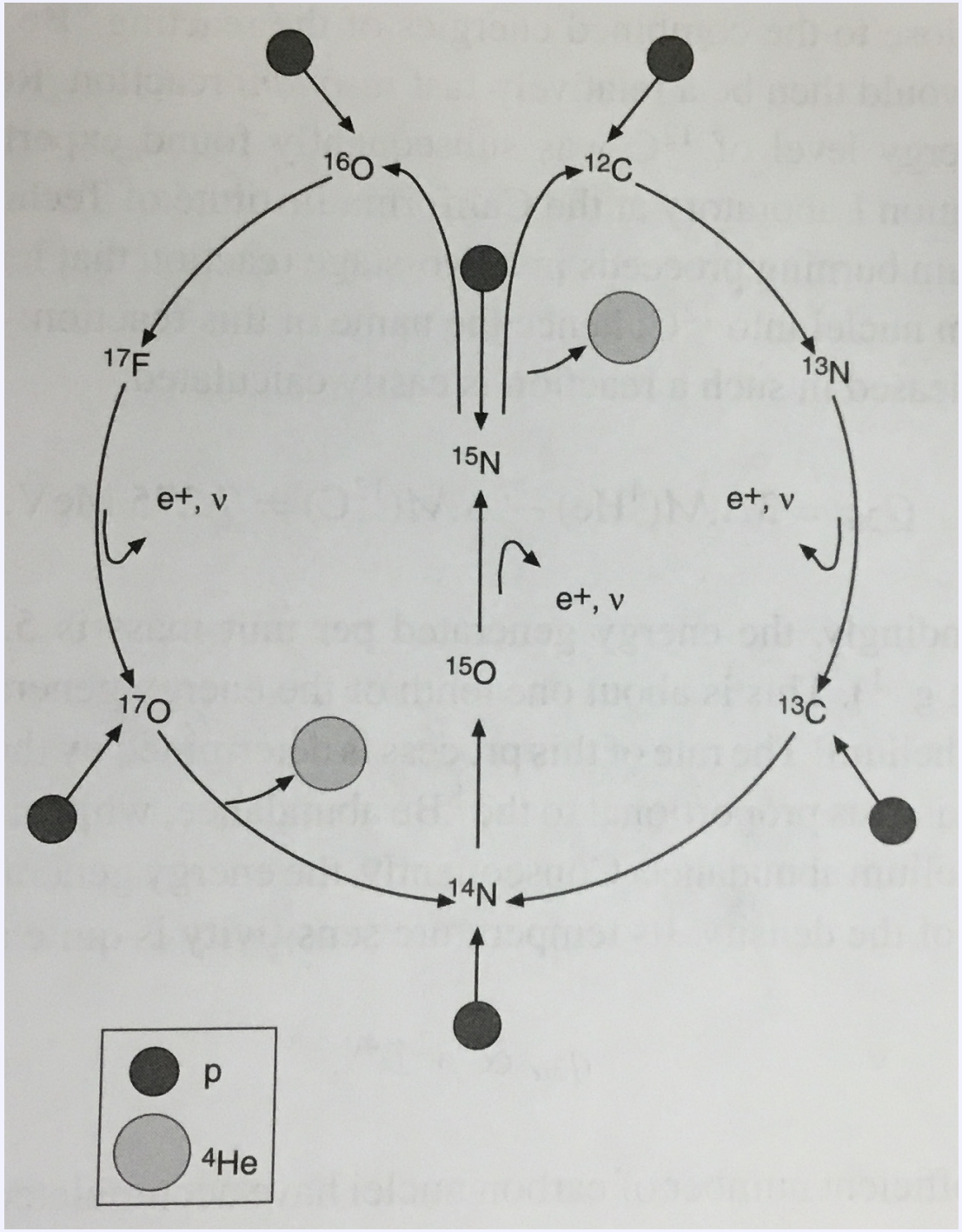
$$^{14}_{7}\mathrm{N} + ^{1}_{1}\mathrm{H} \rightarrow ^{15}_{8}\mathrm{O} + \gamma$$

- Step 5 - average reaction time ~ 3 min:

$$^{15}_{8}\mathrm{O} \rightarrow ^{15}_{7}\mathrm{N} + e^{+} + \nu$$

- Step 6 - average reaction time ~ 30 yr:

$$^{15}_{7}\mathrm{N} + ^{1}_{1}\mathrm{H} \rightarrow ^{12}_{6}\mathrm{C} + ^{4}_{2}\mathrm{He}$$



(Figure credit: Prialnik)

CNO cycle

- Effective energy release is $Q_{\rm eff} = 25.02\,\mathrm{MeV}$
- Total energy generation rate:

$$\epsilon_{CNO} \propto \frac{n_H\, n_{CNO}\, T^{18}}{\rho} \propto X\, Z_{CNO}\, \rho\, T^{18}$$

- This hydrogen fusion more common in hotter stars [O,B,A type stars]
- This cyclic reaction was ineffective for the first generation of stars as it requires an already sufficient heavier element abundance

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Helium burning

- While helium is already used in the hydrogen process, actual helium burning does not start yet then.
- This is due to atomic nuclei with atomic weight 8 not being stable (e.g. ${}^{8}_{4}\mathrm{Be}$) and immediately decaying.
- Instead helium burning happens through the triple-alpha process, needing temperatures around 10^8 K

$$ {}^{4}_{2}\mathrm{He}({}^{4}_{2}\mathrm{He},)^{8}_{4}\mathrm{Be}({}^{4}_{2}\mathrm{He},\gamma)^{12}_{6}\mathrm{C} $$

- Note this needs enough He around to react with the unstable Be before it decays.
- Total energy generation rate:

$$ \epsilon_{3\alpha} \propto Y^3 \, \rho^2 \, T^{30} $$

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Advanced burning

- Once helium is exhausted, the next element to start burning is carbon
- This happens at $T \sim 5 - 10 \times 10^8\,\mathrm{K}$
- Note that core burning for energy generation thus does not happen for all elements
- Next burning phase, based on their Coulomb barrier, would be oxygen, if temperatures reach $10^9\,\mathrm{K}$
- However, at those temperatures, photon energies go so high that they break up elements again (photodisintegration) allowing for Neon (product of carbon fusion) to be burned before oxygen - remember lecture 2.

Nuclear fusion in massive stars

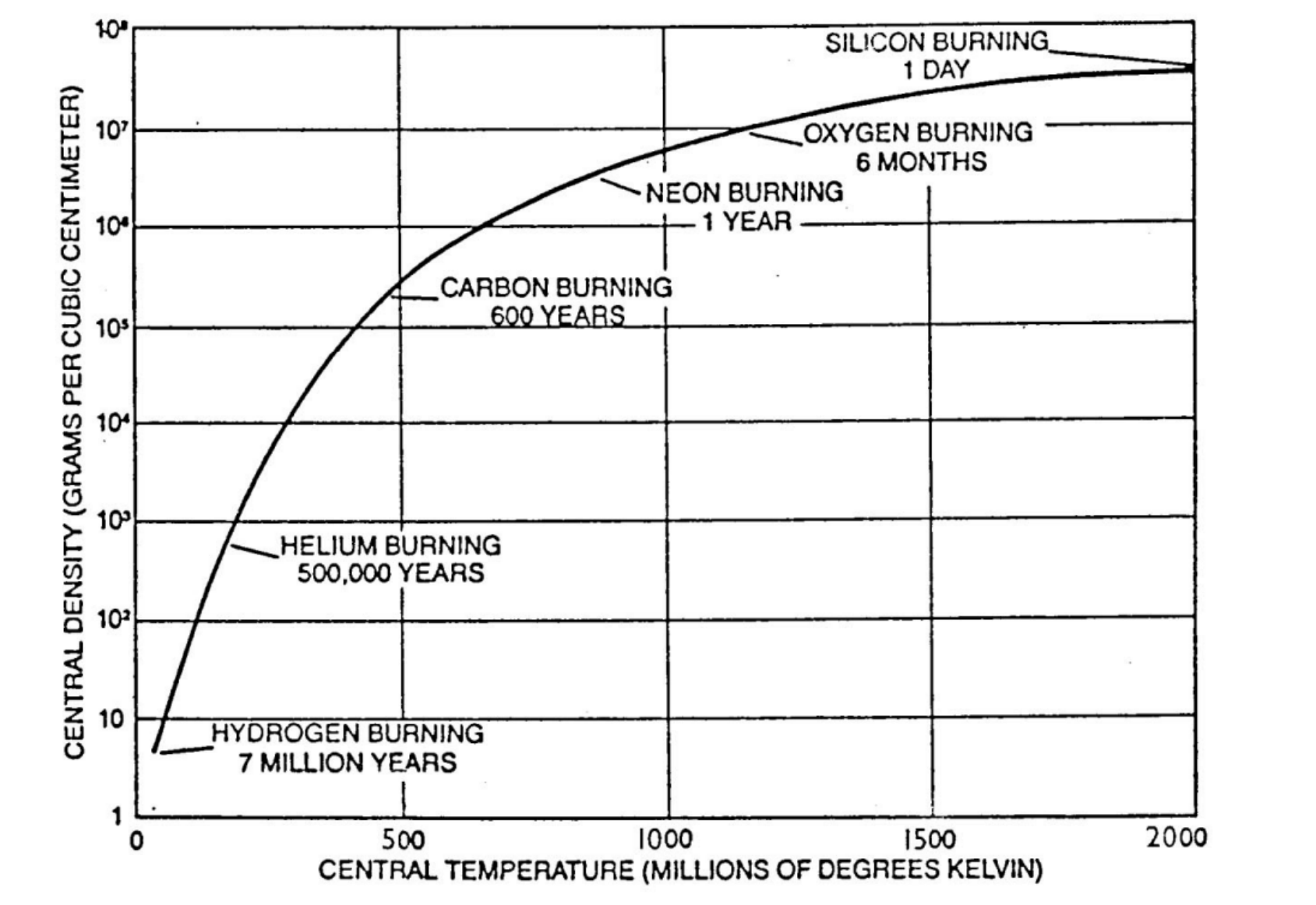

Image credit: JCD

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All equations of stellar structure

$$\text{Mass equation: } \frac{dm}{dr} = 4\pi r^2 \rho$$

$$\text{Hydrostatic equilibrium: } \frac{dP}{dr} = -\frac{G\, m\rho}{r^2}$$

$$\text{Energy generation: } \frac{dL}{dr} = 4\pi\, r^2 \left[\rho\epsilon - \rho \frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho}\frac{d\rho}{dt}\right]$$

$$\text{Radiation transport: } \frac{dT}{dr} = -\frac{3\,\kappa\rho\, L}{16\pi\, a\, c\, r^2\, T^3}$$

$$\text{Convection transport: } \left(\frac{dT}{dr}\right)_{\text{ad}} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dr}$$

$$\text{Chemical evolution: } \frac{dX}{dt} = r_X \text{ etc.}$$

All equations of stellar structure - wrt mass

$$\text{Mass equation: } \frac{dr}{dm} = \frac{1}{4\pi r^2 \rho}$$

$$\text{Hydrostatic equilibrium: } \frac{dP}{dm} = -\frac{G\,m}{4\,\pi\,r^4}$$

$$\text{Energy generation: } \frac{dL}{dm} = \epsilon - \frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho^2}\frac{d\rho}{dt}$$

$$\text{Radiation transport: } \frac{dT}{dm} = -\frac{3\,\kappa\,L}{64\pi^2\,a\,c\,r^4\,T^3}$$

$$\text{Convection transport: } \left(\frac{dT}{dm}\right)_{\text{ad}} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dm}$$

$$\text{Chemical evolution: } \frac{dX}{dt} = r_X \text{ etc.}$$

Additional essential relations

- The dependent variables in these equations are m or r, P, L, T, X [and other chemical mass fractions]
- Checking the right-hand-side of the equations, we thus need expressions for $\rho, \epsilon, u, \kappa, \Gamma_2, r_X$ in terms of pressure, temperature and chemical compostion.
- We've seen these throughout the lectures, eg

$$\kappa \approx \kappa_0 \rho^{\lambda}\, T^{-\nu}$$

$$P = \frac{\rho\, k\, T}{\mu\, m_u} \text{ [for an ideal gas]}$$

$$\mu^{-1} = \sum_j X_j \frac{Z_j + 1}{A_j}$$

$$\epsilon \approx \epsilon_0\, \rho\, T^{\nu}$$

Boundary conditions

- At the centre of the star, it holds that

$$r = m = L = 0$$

- Define the surface of the star as the point where $T = T_{\text{eff}}$. We then have the conditions that at that point

$$m = M_*, r = R_*$$

$$L = L_* = 4\pi\sigma\, R_*^2\, T_{\text{eff}}^4$$

- Note that at this surface, pressure is not zero. It is defined through the use of atmospheric models [see JCD 5.4]

Initial conditions

- Need a model of the starting conditions in star since energy generation as well as chemical evolution have a time dependence
- It turns out that choosing these initial conditions do not have a large effect on later evolution
- Starting from a chemically homogeneous star in thermal equilibrium is the easiest initial condition

Solving the equations of stellar structure and evolution

- Not analytically possible if done properly
- Advanced numerical techniques employed

Summary

- Hydrogen burning happens through pp-chains or CNO cycle
- Helium burning happens through the triple alpha process
- More advanced burning follows burning C, Ne, and O, in that order
- Equations of stellar structure and evolution to describe the variables m or r, P, L, T, X
- Additional relations, boundary conditions and initial conditions then used to numerically solve them

NEXT LECTURE: Early stages star formation, Jeans mass, approach to main sequence